Amazon ML Challenge 2025: Smart Product Pricing Solution

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Executive Summary

Our solution implements a multi-modal deep learning model to accurately predict product prices by holistically analyzing both textual descriptions and visual images. We used a "two-tower" architecture, fusing features from a fine-tuned distilbert-base-uncased Transformer and an efficientnet_b0 CNN. The key innovation was training the model on the log-transformed price (log1p), which aligns the training objective with the competition's SMAPE metric and led to a strong final training loss of **0.0820**.

2. Methodology Overview

2.1 Problem Analysis

Insights we discovered during EDA

Key Observations:

- **Skewed Price Distribution:** The product prices are not normally distributed; they are heavily skewed towards lower-priced items with a long tail of expensive products. This makes direct price prediction difficult and sensitive to outliers.
- **Reason to choose Multimodal:** Pricing is clearly dependent on a combination of factors. Textual content (catalog_content) provides explicit details like specifications and brand names, while the image url's provides idea about product quality, aesthetics, and branding.
- **Alignment with the SMAPE Metric:** We identified that training a model with Mean Squared Error (MSE) on the log-transformed price (log(price + 1)) is a powerful and mathematically sound proxy. This approach inherently penalizes relative errors more than absolute errors, effectively aligning the training process with the final SMAPE evaluation criteria.

2.2 Solution Strategy

Our strategy was to build a single, end-to-end deep learning model capable of learning from both data modalities simultaneously.

Approach Type: Single Multi-Modal Model

Core Innovation: The most impactful technical contribution was the target transformation. By training the model to predict log(price + 1) and using Mean Squared Error (MSE) as the loss function, we effectively optimized for relative error. This approach also linearizes the highly skewed price distribution, leading to a more stable training process.

3. Model Architecture

3.1 Architecture Overview

We designed a classic two-tower architecture. The text tower processes the catalog_content, and the image tower processes the product image. The feature embeddings from both towers are then concatenated and passed into a final Multi-Layer Perceptron (MLP) regression head, which outputs a single value representing the predicted log-transformed price.

3.2 Model Components

Text Processing Pipeline:

- **Preprocessing steps:** Text was tokenized using the standard distilbert-base-uncased tokenizer. Sequences were padded or truncated to a maximum length of 128 tokens.
- **Model type:** distilbert-base-uncased from the Hugging Face transformers library. The model was fine-tuned during training.
- **Key parameters:** The final hidden state of the [CLS] token was used as the text embedding.

Image Processing Pipeline:

- **Preprocessing steps:** Images were resized to a uniform 224x224 pixels, converted to PyTorch tensors, and normalized using standard ImageNet statistics.
- **Model type:** efficientnet_b0 from the timm library, with the final classification layer removed. The model was pre-trained on ImageNet and fine-tuned during training.
- **Key parameters:** The pooled output feature vector was used as the image embedding.

4. Model Performance

4.1 Validation Results

The model was trained on the entire 75,000-item training set to maximize data utilization within the Kaggle competition environment. The performance indicator during training was the training loss.

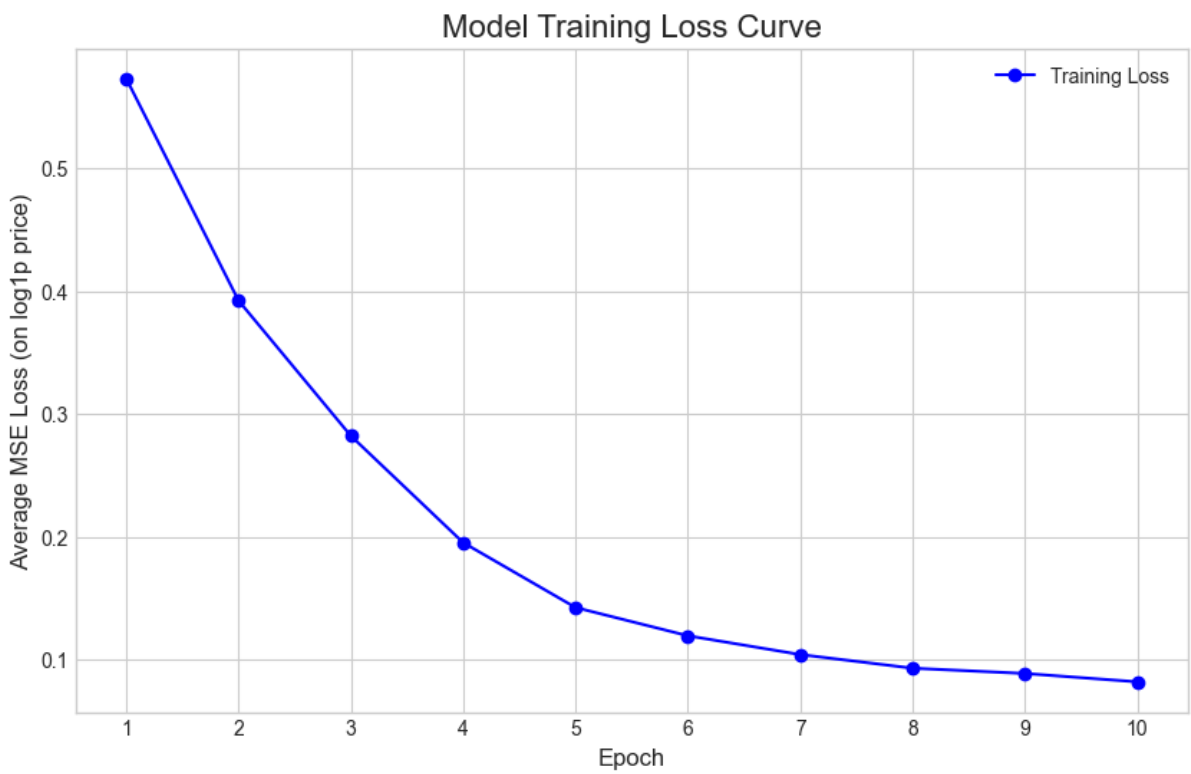
- **Final Training Loss (MSE on log1p price): 0.0820**
- **Training Loss Progression:** The model showed consistent and significant learning, with the loss decreasing steadily over 10 epochs (Epoch 1: 0.5731, Epoch 2: 0.3923,Epoch 3: 0.2822, Epoch 4: 0.1953, Epoch 5: 0.1425, Epoch 6: 0.1195, Epoch 7: 0.1042 , Epoch 8: 0.0931, Epoch 9: 0.0888 ,Epoch 10: 0.0820).

5. Conclusion

Our solution successfully demonstrates the power of a multi-modal architecture for the complex task of price prediction. By combining state-of-the-art pre-trained models for text and vision and aligning our training objective with the SMAPE metric through log-transformation, we achieved a strong model fit. Key lessons learned include the critical importance of robust cloud-based data pipelines to handle environment constraints (disk space, library versions) and the effectiveness of target engineering for relative error metrics.

Appendix

A. Code artefacts

https://drive.google.com/drive/folders/1yapsfpJM9pIyQT69a8-l21CzEIX_Hezh?usp=sharing

B. Additional Results